

Corvus installation instructions

Requirements:

1. FEFF10: <https://times-software.github.io/feff10/>
2. Python: <https://www.python.org/downloads/>

Warning: This installation is fairly large due to the use of pymatgen. It can take a while to download and install everything. Expect 10 to 20 minutes for the installation process to complete.

Note: The following download instructions will make a venv python environment in your home directory, i.e.,

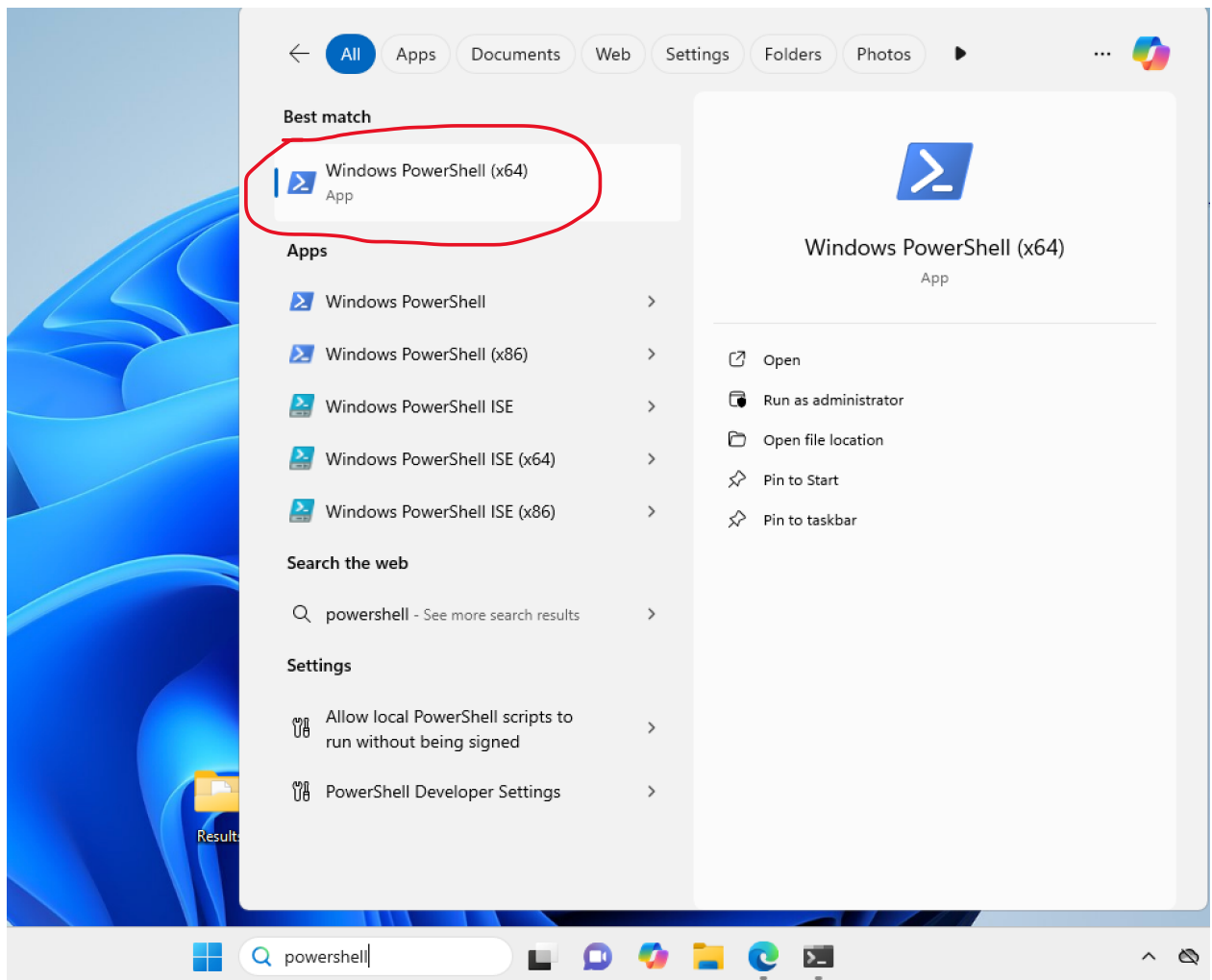
`your_home/.venv/Corvus`

and install `corvus` in that directory. If you do not want to use a venv environment (for example if you use anaconda) you should perform a manual install as detailed below.

Windows:

1. Download the latest release version of `corvus`:
<https://github.com/times-software/corvus/releases/latest>
2. Unzip the archive.

3. Open a powershell terminal



4. Type

```
cd ~\Downloads
```

```
cd Corvus-[latest version]\Corvus-[latest version]
```

where [latest version] signifies the version that you downloaded, e.g., Corvus-1.5.0

5. Type

```
.\install_w_venv.ps1
```

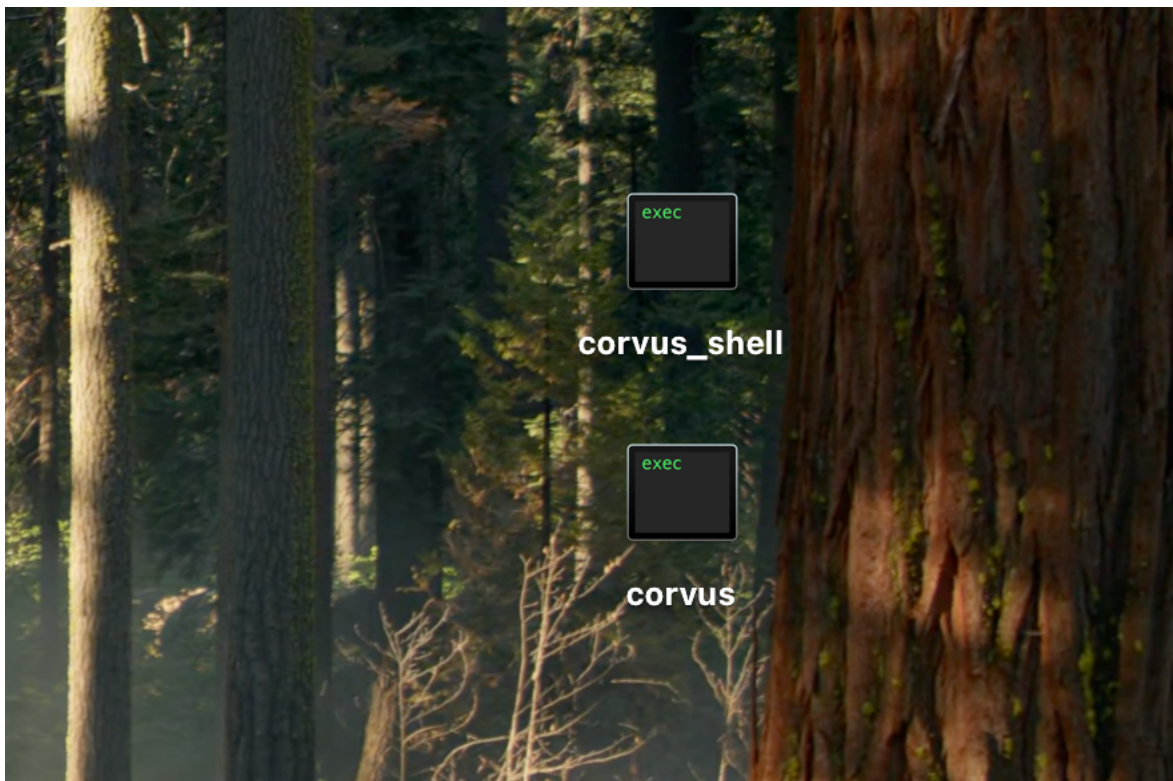
and follow the instructions to install corvus.

6. There will now be a shortcut on your desktop named “corvus_shell” that will open an interactive corvus terminal.

7. There will also be a shortcut on your desktop named “corvus” that will open the corvus graphical user interface.

Mac OS:

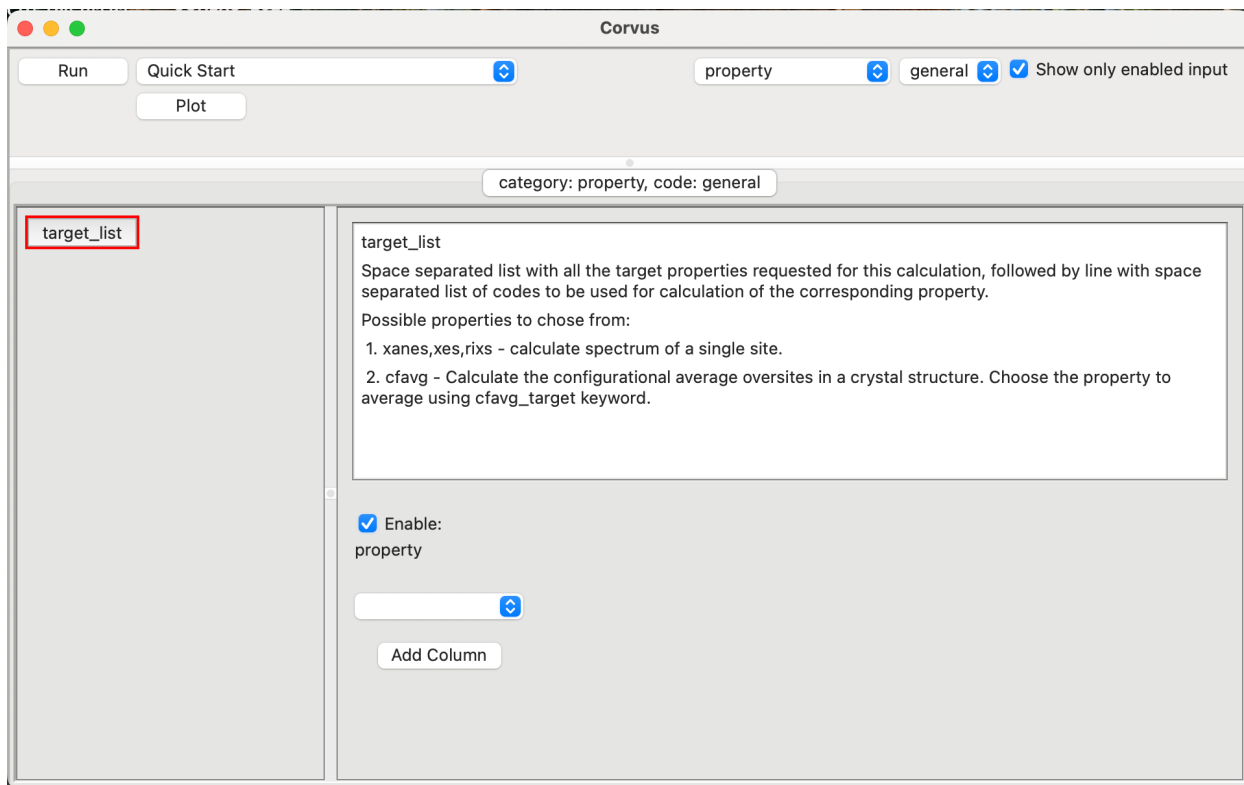
1. Download the latest release version of corvus:
<https://github.com/times-software/corvus/releases/latest>
2. Unzip the archive.
3. Open a terminal: “Applications -> Utilities -> Terminal
4. Type the following command in the terminal to change directory to the downloaded corvus directory:
`cd ~/Downloads/Corvus-[latest version]/`
where [latest version] signifies the version that you downloaded, e.g., Corvus-1.5.0
5. Type:
`source install_w_venv.sh`
and follow the instructions.
6. There will now be several executable scripts on your desktop. Double click to start corvus:



- a. **corvus_shell** – open a shell to run corvus in the terminal
- To test the installation, you can type:
- ```
run-corvus --version
```

```
jjkas — corvus_shell — zsh -i — 80x24
Last login: Sat Sep 20 10:43:17 on ttys002
/Users/jjkas/Desktop/corvus_shell ; exit;
jjkas@Joshuas-MacBook-Pro-9 ~ % /Users/jjkas/Desktop/corvus_shell ; exit;
[(Corvus) jjkas@Joshuas-MacBook-Pro-9 ~ % run-corvus --version]
#####
#
Corvus version 1.1.5
#
#
#####
(Corvus) jjkas@Joshuas-MacBook-Pro-9 ~ %
```

**b. corvus – open the corvus graphical user interface.**



**Note: Sometimes the above window does not come to the front on MacOS. If**

you don't see it look behind other windows or for the python icon in your toolbar.

### Manual installation:

If you do not want to use venv environments, or the above installations don't work for your system, use the manual installation instructions below.

1. Download the latest release version of corvus:  
<https://github.com/times-software/corvus/releases/latest>
2. Unzip the archive.
3. Go into the expanded archive directory. Inside it you should see a setup.py file.
4. Optional: create a python environment and activate it to isolate this installation from your other python projects.
5. Install with pip.  
`pip3 install .`
6. If you want the corvus graphical user interface, download it at:  
<https://github.com/times-software/SciGUI/archive/refs/heads/main.zip>
7. Unzip the downloaded file SciGUI-main.zip.
8. Use the terminal to install it. If you created a python environment, make sure it is activated before using pip to install.  
`cd ~/Downloads/SciGUI-main`  
**Activate your environment if you made one**  
`pip3 install .`
9. Now in the future, to start corvus:
  - a. Open a terminal
  - b. Activate your environment if you chose to make one
  - c. Test corvus  
`run-corvus --version`
  - d. Open the corvus GUI  
`corvus`